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→ Appⁿ: ① dimensionality reduction

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Eigen Vectors & Eigen Values

⇒ Eigen value (λ): It's a scalar that indicates how much an eigen vector is stretched or compressed during linear transformation.

⇒ Eigen vector (V): A non-zero vector that only changes in scale (not direction) when a linear transformation is applied.

⇒ For square matrix 'A', an eigen vector & the corresponding eigen value ' λ ' satisfy the equation:

$$AV = \lambda V$$

eg,

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

∴ eigen value:

$$\det(A - \lambda I) = 0$$

$$= \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{bmatrix}$$

$$= (4-\lambda)(3-\lambda) - 2 \times 1$$

$$= \lambda^2 - 7\lambda + 10 = \lambda^2 - 5\lambda - 2\lambda + 10$$

$$= \lambda(\lambda-5) - 2(\lambda-5) = (\lambda-5)(\lambda-2)$$

$$\therefore \lambda_1 = 5, \lambda_2 = 2$$

→ eigenvalues of 'A'

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∴ Eigenvector:

$$\lambda(A - \lambda I)v = 0$$

for $\lambda = 5$

$$A - 5I = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix}$$

$$\therefore \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\Rightarrow -x + y = 0 \Rightarrow x = y$$

for $\lambda_2 = 2$

$$A - 2I = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$$

$$\therefore \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

$$\Rightarrow 2x + y = 0 \Rightarrow y = -2x$$

So, for $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ and $\lambda_1 = 5, \lambda_2 = 2$

for $\lambda_1 = 5$ $v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\lambda_2 = 2$ $v_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$



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